GeoDP: Geometry-Aware Differential Privacy for Face Template Protection

Abstract

Deep face recognition systems are based on compact biometric embeddings that are vulnerable to template reconstruction, cross-database linkage, and membership inference attacks.Differential privacy (DP) offers rigorous privacy guarantees and has recently been explored for biometric template protection through both image- and embedding-level perturbations. However, image-level approaches often degrade the discriminative quality of facial representations, whereas existing embedding-level methods typically perturb feature dimensions uniformly, disregarding the highly anisotropic geometry of deep face embeddings. Consequently, current DP-based approaches exhibit a suboptimal privacy–utility trade-off under practical privacy budgets. We present *GeoDP*, a geometry-aware differential privacy framework that protects deep face templates directly in the embedding space. GeoDP learns a Fisher discriminative subspace from a representative population and constructs an adaptive covariance whose variance is determined by the discriminative importance of each Fisher direction. It further bounds perturbations in weakly discriminative directions to prevent unnecessary distortion while preserving the full geometry-aware Gaussian perturbation in the dominant discriminative subspace. The protected templates are subsequently ℓ_2-normalized, making GeoDP directly compatible with existing cosine-similarity-based face recognition systems without modifying the recognition pipeline.

The operating parameters are selected through multi-objective Pareto optimization, and the framework is evaluated on CASIA-WebFace, VGGFace2, LFW, and FEI under a white-box threat model. At the Pareto-optimal operating point ($\epsilon = 12$), GeoDP achieves **93.67%** Rank-1 identification accuracy on CASIA-WebFace, substantially outperforming representative image-level and isotropic embedding-level differential privacy baselines. GeoDP further limits its reconstruction attacks to an SSIM of approximately **0.35**, reduces membership inference performance toward random guessing, and achieves low empirical linkability. These results demonstrate that leveraging the intrinsic geometry of deep face embeddings substantially improves the privacy-utility trade-off achievable by differential privacy for face template protection.

Keywords

Differential Privacy, Biometric Template Protection, Face Recognition, Face Reconstruction, FisherFaces, Membership Inference Attack, Unlinkability

1 Introduction

Deep face recognition systems have become one of the most widely adopted biometric technologies due to their high recognition accuracy, robustness, and scalability [18, 45]. They are widely used in smartphone authentication, border control, financial services, healthcare, surveillance, digital identity management, and large-scale access control systems. These systems typically store biometric templates as compact deep face embeddings [12, 39], enabling efficient large-scale identity matching. However, unlike passwords or cryptographic keys, biometric templates are inherently linked to an individual's physiological characteristics and cannot be revoked or reissued once compromised. Consequently, protecting stored biometric templates has become a fundamental requirement to ensure the security and privacy of face recognition systems [26, 33].

A compromised facial template may facilitate long-term identity theft, unauthorized authentication, and user impersonation. Recent advances in machine learning have demonstrated that deep face embeddings are vulnerable to template inversion [20], face reconstruction [31], and membership inference attacks [41]. Reconstruction attacks attempt to recover face images from stored embeddings, whereas the membership inference attacks determine whether an individual's biometric data is included in the enrollment database. Moreover, the deterministic nature of face embeddings enables cross-database linkage attacks, allowing adversaries to link templates belonging to the same individual across independent systems, thus violating the unlinkability requirement of biometric template protection [1, 23]. These attacks have motivated the development of face template protection mechanisms that preserve recognition utility while preventing the leakage of sensitive biometric information.

Various approaches have been proposed to protect biometric data, including biometric cryptosystems [8, 14, 27], cancelable biometrics [9, 37, 38], privacy-preserving computation techniques [3, 47], and deep learning-based protection methods [22, 28, 30]. Despite significant progress, existing approaches either sacrifice recognition accuracy, require specialized matching protocols, or provide limited formal privacy guarantees, making their deployment challenging in large-scale face recognition systems. Differential Privacy (DP) has emerged as a mathematically rigorous framework to limit the information that can be inferred about an individual from the protected template while providing quantifiable privacy guarantees [15, 17]. Existing differential privacy methods protect facial representations by injecting calibrated random perturbations either before feature extraction (image-domain perturbation) or after feature extraction (embedding-domain perturbation) [5, 19, 32, 35, 46]. Nevertheless, effectively applying differential privacy to deep face representations while maintaining high recognition performance remains a significant challenge.

1.1 Motivation

Existing differential privacy approaches introduce privacy protection at different stages of the face recognition pipeline, including input face images, intermediate feature representations, and biometric embeddings [5, 19, 32, 35, 46]. Although these methods provide formal privacy guarantees, they continue to exhibit a suboptimal privacy–utility trade-off. Figure 1 shows the transition of DP based face template protection from the existing approaches to our proposed approach.

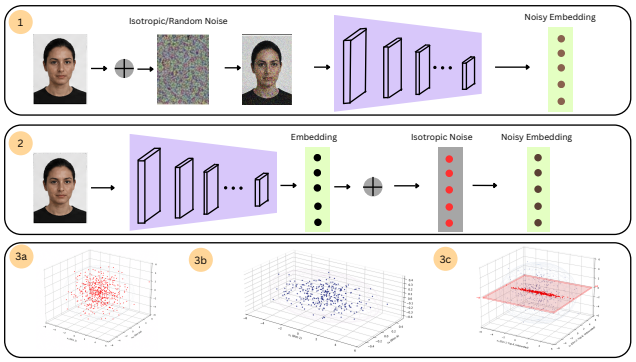


Figure 1: Motivation of the proposed work. Existing differential privacy mechanisms either (1) perturb input images before feature extraction with random or isotropic noise, or (2 – 3*a*) employ isotropic perturbation on biometric embeddings, limiting the achievable privacy–utility trade-off. GeoDP (3*b*) instead performs geometry-aware random perturbation on face embeddings and (3*c*) shows the proposed adaptive GeoDP approach where bounded and unbounded noise is applied to the face embedding, achieving high utility vs privacy tradeoff.

- **Image-level differential privacy perturbs the representation before feature extraction.** Recent approaches achieve differential privacy by perturbing input face images prior to feature extraction [19, 32, 35]. While these methods provide privacy guarantees in the image domain, the injected perturbation propagates through the deep feature extractor, altering identity-discriminative facial characteristics before the biometric template is generated. Since modern face recognition systems ultimately store and match compact face embeddings rather than raw images, protecting the image instead of the stored template often leads to unnecessary degradation of recognition performance while providing no direct control over the privacy characteristics of the generated embedding.
- **Embedding-level isotropic perturbation ignores the discriminative geometry of deep face embeddings.** Existing embedding-level DP mechanisms typically employ isotropic Gaussian perturbation or uniformly scaled noise across all embedding dimensions [5]. Such approaches implicitly assume that every embedding dimension contributes equally to identity discrimination. However, deep face embeddings exhibit a highly anisotropic statistical structure, where different projection directions contribute unequally to recognition performance [2, 12, 29]. Consequently, uniformly allocating the privacy budget across all embedding dimensions excessively perturbs highly discriminative directions while applying the same perturbation to the weakly discriminative dimensions, resulting in an inefficient privacy-utility trade-off.

These observations motivate the proposed *GeoDP* framework, which performs differential privacy directly on the embeddings while exploiting the intrinsic statistical geometry of deep face embeddings. By adaptively allocating perturbation according to the discriminative importance of embedding directions, GeoDP preserves recognition utility while maintaining rigorous differential privacy guarantees.

1.2 Contributions

The main contributions of our work are summarized below.

- **Geometry-Aware Differential Privacy Framework:** We propose **GeoDP**, a novel differential privacy framework that protects face templates directly in the embedding space. Unlike existing image-level or isotropic embedding-level perturbation methods, GeoDP exploits the discriminative geometry of deep face embeddings to allocate privacy noise more effectively while remaining fully compatible with existing cosine-similarity-based face recognition systems.
- **Adaptive Geometry-Aware Perturbation Mechanism:** We introduce an adaptive perturbation mechanism that constructs a geometry-aware covariance from the Fisher discriminative subspace and selectively combines unbounded Gaussian perturbation for highly discriminative embedding directions with bounded Gaussian perturbation for weakly discriminative directions. This design provides stronger perturbation in identity-sensitive directions while minimizing unnecessary distortion in weakly discriminative regions of the embedding space.
- **Multi-objective Privacy-Utility Optimization:** We formulate parameter selection as a constrained multi-objective optimization problem and identify Pareto-optimal operating points that maximize template unlinkability while satisfying a desired recognition accuracy.
- **Theoretical and Experimental Validation:** We theoretically analyze the security properties of GeoDP and establish its differential privacy guarantee. Extensive experiments on CASIA-WebFace, VGGFace2, LFW, and FEI demonstrate that GeoDP consistently achieves a superior privacy-utility trade-off, while providing strong resistance against template reconstruction, membership inference, and cross-database linkage attacks.

2 Models and Settings

This section describes the system architecture, reviews the differential privacy concepts required throughout the paper, and presents the adversarial model considered in this work.

2.1 System Model

The proposed biometric authentication system follows a client-server architecture where the client is assumed to be a genuine or an impostor user. The server is assumed to be honest-but-curious; it correctly executes the authentication protocol but may attempt to infer sensitive information from the stored templates. During enrollment and authentication, the client processes the captured face image using a pre-trained ArcFace network [13] to obtain a 512-dimensional face embedding which is transformed into a protected face template using the proposed GeoDP framework.

The comparisons between two face templates are performed in the protected domain.

2.2 Differential Privacy: Background

Let $\mathbf{x} \in \mathbb{R}^d$ denote the original face embedding extracted from a facial image and $\tilde{\mathbf{x}} = \mathcal{M}(\mathbf{x})$ denote the protected template produced by a randomized mechanism $\mathcal{M}$. The mechanism satisfies (ϵ, δ)-differential privacy [15] if, for any pair of neighboring biometric templates $\mathbf{x}$ and $\mathbf{x}'$ differing by a single biometric record, and for any measurable output set $\mathcal{S}$,

$$\Pr[\mathcal{M}(\mathbf{x}) \in \mathcal{S}] \leq e^{\epsilon} \Pr[\mathcal{M}(\mathbf{x}') \in \mathcal{S}] + \delta, \tag{1}$$

where ϵ controls the privacy loss and δ denotes the probability of relaxing the strict privacy guarantee. A smaller value of ϵ provides stronger privacy protection at the cost of increased perturbation.

The Gaussian mechanism is the most widely adopted approach for achieving (ϵ, δ)-differential privacy in continuous-valued data. Given a query function $f(\cdot)$ with global L_2 sensitivity Δ_2, the released output is generated by perturbing the original biometric template with Gaussian noise, $\mathcal{M}(\mathbf{x}) = f(\mathbf{x}) + \mathcal{N}(\mathbf{0}, \Sigma)$, where Σ denotes the covariance matrix of the injected perturbation. Conventional Gaussian mechanisms typically assume an isotropic covariance matrix given as $\Sigma = \sigma^2 I$, which allocates identical perturbation variance to every embedding dimension. The proposed framework differs from this assumption by constructing a geometry-aware covariance, described in Section 3.

2.3 Threat Model

We consider an adversary who gains access to one or more protected biometric templates stored on the authentication server. The adversary is assumed to have complete knowledge of the face recognition model, proposed protection algorithm, and some auxiliary knowledge, such as a collection of original and protected template pairs; but has no access to any secret parameters used by the system. The adversary aims to compromise the privacy of enrolled users through the following attacks.

- **Face Reconstruction Attack:** The adversary aims to recover the original biometric information from a protected template [11, 40]. We assume the adversary has access to multiple pairs of original and protected face embeddings. Using these samples, the adversary trains an inversion model to approximate the reverse mapping from protected templates to the original embedding space, with the ultimate objective of reconstructing a recognizable facial image. Such attacks violate the irreversibility requirement of ISO/IEC 24745 [1] and may reveal sensitive biometric information.
- **Cross-Database Linkage Attack:** The adversary obtains protected biometric templates belonging to the same users from multiple independent databases and attempts to determine whether two templates originate from the same individual. Deterministic or highly correlated template representations can be linked using similarity measures such as cosine similarity or Euclidean distance, enabling cross-application user profiling and violating the unlinkability requirement of biometric template protection.
- **Membership Inference Attack:** The adversary tries to determine whether an individual's biometric data was included in the model's training set (CASIA Members) or or it is a non-member. By exploiting statistical differences between templates corresponding to enrolled and non-enrolled users, the adversary infers the enrollment status of the target without explicitly reconstructing the underlying biometric information, potentially revealing participation in privacy-sensitive applications.

3 Proposed GeoDP Framework

The proposed GeoDP framework consists of two sequential phases: (i) an offline geometry learning stage that estimates the discriminative structure of the embedding space from a representative population, and (ii) an online template protection stage that generates privacy-preserving biometric templates through geometry-aware differential privacy. Figure 2 illustrates the entire pipeline.

3.1 Offline Geometry Learning

The proposed geometry-aware perturbation requires prior knowledge of how identity information is distributed within the face embedding space. Since this structure depends on the statistical characteristics of a representative population rather than individual users, it can be estimated once during system initialization. Hence, the online protection stage incurs only minimal computational overhead.

Let $\mathcal{X} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ denote a labeled training set, where $\mathbf{x}_i \in \mathbb{R}^{512}$ is the ArcFace embedding extracted from the i-th training image and y_i denotes its identity label. To estimate the discriminative structure of the embedding space, we employ the FisherFace framework, which combines Principal Component Analysis (PCA) with Linear Discriminant Analysis (LDA) [2]. PCA first projects the high-dimensional embeddings onto a lower-dimensional subspace, removing redundant correlations and alleviating the small sample size problem. Then, LDA learns a set of orthogonal projection directions that maximize inter-class variations while minimizing intra-class variations.

The resulting transformation is represented by the Fisher projection matrix $W_F \in \mathbb{R}^{512\times128}$, which projects every ArcFace embedding into a 128−dimensional discriminative subspace,

$$\mathbf{z} = W_F^T \mathbf{x}.$$

The Fisher decomposition also produces an ordered eigenvalue spectrum

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{128} > 0,$$

where each eigenvalue reflects the relative discriminative importance of its corresponding projection direction. Larger eigenvalues indicate directions with stronger identity-discriminative information, whereas smaller eigenvalues correspond to weaker discriminative directions.

Unlike conventional FisherFace-based recognition systems, the learned subspace is not used for recognition itself. Instead, it provides a geometric prior that guides the proposed differential privacy mechanism. Accordingly, the offline phase produces two model parameters- the Fisher projection matrix W_F, which maps every

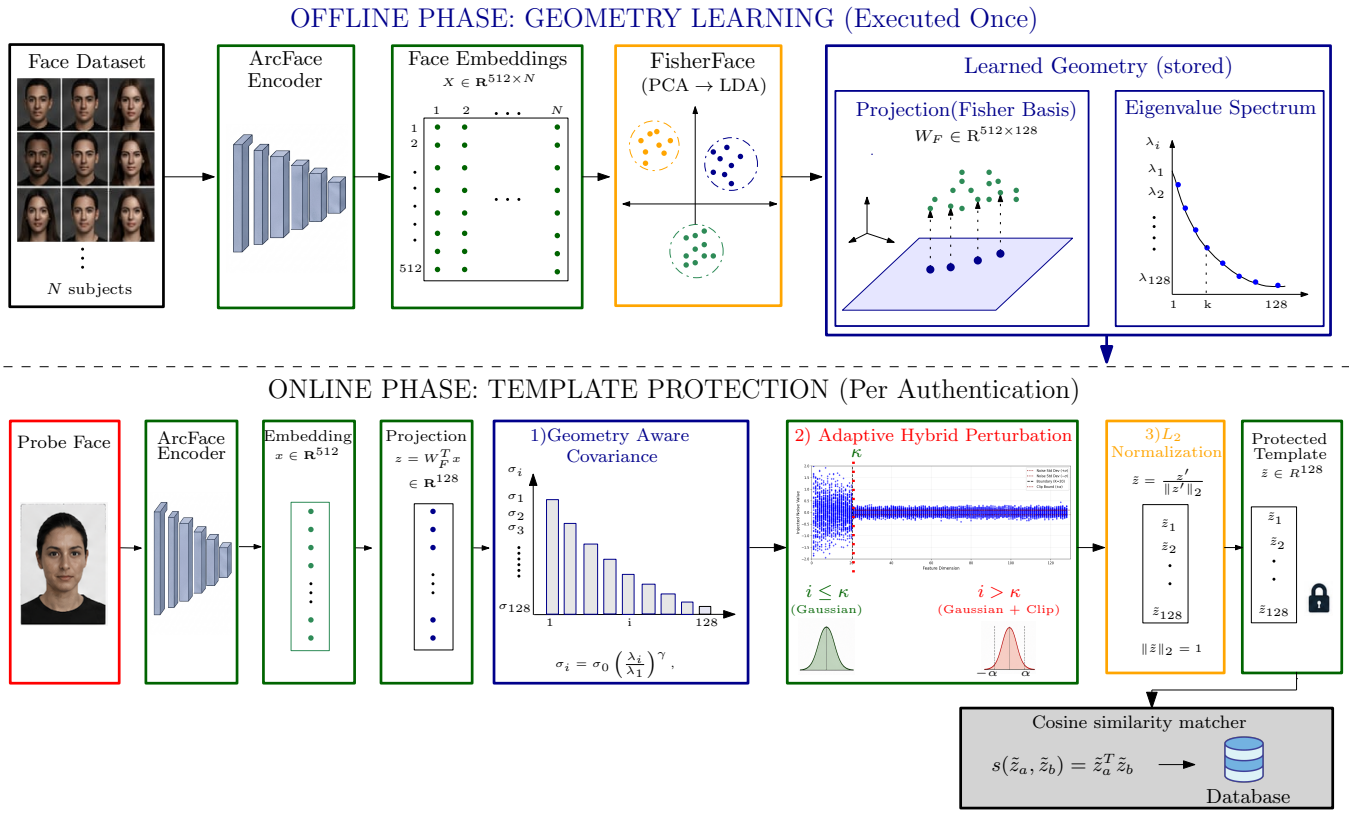


Figure 2: Proposed model for Geometry-Aware Differential Privacy for face template protection.

face embedding into the learned discriminative coordinate system, and the corresponding eigenvalue spectrum $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_{128})$, which encodes the identity-discriminative importance of each projection direction. These parameters are stored as fixed system parameters and subsequently reused during the online phase.

3.2 Online Phase: Template Protection

During enrollment or authentication, each input face image is first transformed into a 512−dimensional ArcFace embedding. The embedding is projected into the Fisher discriminative subspace learned during the offline phase. This projection represents the embedding in a coordinate system where different feature directions contribute unequally to identity discrimination. Our proposed geometry-aware differential privacy mechanism exploits this statistical geometry to construct an adaptive perturbation whose covariance varies across the Fisher directions. Finally, the perturbed embedding is normalized onto the unit hypersphere and stored as the protected biometric template.

3.2.1 Projection into the Fisher Discriminative Subspace. Let

$$\mathbf{x} = [x_1, x_2, \ldots, x_{512}]^T \in \mathbb{R}^{512}$$

denote the normalized ArcFace embedding extracted from an input face image. Using the Fisher projection matrix learned in the offline phase,

$$W_F = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_{128} \end{bmatrix} \in \mathbb{R}^{512\times 128}.$$

where

$$\mathcal{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{128}\},$$

represents a set of orthogonal discriminative Fisher directions, each associated with the eigenvalue λ_i.

Using the learned projection matrix, the ArcFace embedding is projected onto the Fisher discriminative subspace as

$$\mathbf{z} = W_F^T \mathbf{x} = \begin{bmatrix} \mathbf{v}_1^T \mathbf{x} \\ \mathbf{v}_2^T \mathbf{x} \\ \vdots \\ \mathbf{v}_{128}^T \mathbf{x} \end{bmatrix} = [z_1, z_2, \ldots, z_{128}]^T, \tag{2}$$

where

$$z_i = \mathbf{v}_i^T \mathbf{x}$$

denotes the projection coefficient along the i-th Fisher direction.

Thus, the projected representation allows the perturbation strength to be allocated according to the statistical importance of each Fisher direction rather than uniformly across all embedding dimensions.

3.2.2 Geometry Aware Noise Perturbation. To perturb the face template while preserving its biometric utility, traditional differential privacy approaches generally employ isotropic Gaussian noise, where identical noise variance is assigned to every embedding dimension [6] ($\sigma_i = \sigma_0$). It may unnecessarily distort highly discriminative components while providing limited additional privacy in insensitive information directions. To address this limitation,

we construct a geometry-aware covariance matrix in which the noise variance assigned to each Fisher direction is determined by its corresponding eigenvalue.

Let the differential privacy budget be specified by (ϵ, δ). Following the Gaussian mechanism, the baseline noise scale is computed as [15, 17]

$$\sigma_0 = \frac{\Delta_2 \sqrt{2\ln(1.25/\delta)}}{\epsilon}, \tag{3}$$

where δ is the privacy failure probability, allowing a very small probability that the privacy guarantee is exceeded. Δ_2 denotes the global L_2 sensitivity of the normalized face embeddings. The parameter ϵ controls the privacy-utility trade-off of the proposed mechanism. Smaller values of ϵ increase the baseline noise variance, thereby providing stronger privacy protection but leads to greater distortion in the protected template. Conversely, larger values of ϵ reduce the injected noise, preserving recognition performance while providing comparatively weaker privacy guarantees.

The larger eigenvalues correspond to directions containing stronger identity-discriminative information, consequently, they receive proportionally stronger noise than weakly discriminative directions in the embeddings. Accordingly, the standard deviation associated with the i-th Fisher direction is defined as

$$\sigma_i = \sigma_0 \left(\frac{\lambda_i}{\lambda_1}\right)^{\gamma}, \tag{4}$$

where λ_1 denotes the largest Fisher eigenvalue and $\gamma > 0$ is the geometry adaptation parameter that controls the influence of the Fisher eigenspectrum on the noise allocation. Since

$$0 < \frac{\lambda_i}{\lambda_1} \leq 1,$$

increasing γ progressively suppresses the noise variance assigned to lower-eigenvalue directions while leaving the dominant direction (λ_1) unchanged. As a result, the injected Gaussian noise introduces less distortion into the protected embedding while preserving the relative geometry-aware allocation of noise across the dimensions. It allows the proposed framework to better balance recognition performance and privacy protection.

The resulting covariance matrix is therefore

$$\Sigma = \sigma_0^2 \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & \left(\frac{\lambda_2}{\lambda_1}\right)^{2\gamma} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \left(\frac{\lambda_{128}}{\lambda_1}\right)^{2\gamma} \end{bmatrix}, \tag{5}$$

The perturbation vector is sampled from the multivariate Gaussian distribution

$$\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \Sigma), \tag{6}$$

which assigns an independent geometry-aware noise to each dimension. The resulting perturbed embedding is obtained as

$$\tilde{\mathbf{z}} = \mathbf{z} + \mathbf{n}. \tag{7}$$

3.2.3 Geometry Aware Adaptive Hybrid Perturbation. Although the geometry aware covariance allocates noise according to the discriminative importance of each Fisher direction, the unbounded support of the Gaussian distribution may still produce large noise realizations. Such perturbations are particularly undesirable in weakly discriminative directions, where they provide little additional privacy while unnecessarily degrading recognition performance.

To mitigate this effect, we partition the Fisher subspace into two complementary regions using a threshold index $1 \leq k \leq 128$. The leading k directions capture the majority of the identity-discriminative information. Therefore, they retain the full geometry-aware Gaussian noise. The remaining directions receive bounded Gaussian noise to suppress excessive distortion. Accordingly, for each dimension,

$$n_i \sim \mathcal{N}(0, \sigma_i^2), \qquad i = 1, \ldots, 128, \tag{8}$$

where σ_i is computed according to Eq. (4). The protected component is then obtained as

$$\tilde{z}_i = \begin{cases} z_i + n_i, & i \leq k, \\ z_i + \hat{n}_i, & i > k, \end{cases} \tag{9}$$

where

$$\hat{n}_i = \mathrm{clip}(n_i, -\alpha, \alpha), \tag{10}$$

The clipping operation bounds the sampled Gaussian noise to the interval $[-\alpha, \alpha]$, preventing excessively large perturbations in weakly discriminative directions. The clipping threshold α denotes the maximum magnitude of noise injected in the weakly discriminative subspace. A smaller value of α limits large Gaussian perturbations. The clipping operation is defined as

$$\mathrm{clip}(n_i, -\alpha, \alpha) = \begin{cases} -\alpha, & n_i < -\alpha, \\ n_i, & -\alpha \leq n_i \leq \alpha, \\ \alpha, & n_i > \alpha. \end{cases} \tag{11}$$

The resulting protected embedding is

$$\tilde{\mathbf{z}} = [\tilde{z}_1, \tilde{z}_2, \ldots, \tilde{z}_{128}]^T. \tag{12}$$

Figure 3 illustrates the resulting hybrid noise profile. The dominant Fisher directions ($i \leq k$) receive full geometry-aware Gaussian noise, whereas the remaining directions ($i > k$) employ bounded perturbation to suppress unnecessary distortion while preserving the statistical structure of the protected embedding. The parameters of the proposed framework, including the privacy budget ϵ, geometry adaptation parameter γ, hybrid partition index k, and clipping threshold α are selected through an offline Pareto optimization process to achieve an effective balance between recognition performance and privacy protection as described in Section 4.

The learned Fisher projection matrix W_F, the eigenvalue spectrum Λ, and the associated geometry-aware covariance matrix are considered public parameters. Consistent with the standard differential privacy threat model, the perturbation mechanism is assumed to be fully known to the adversary. Privacy is preserved through independently sampled Gaussian noise for each enrollment or authentication instance, making the realized perturbation statistically unpredictable.

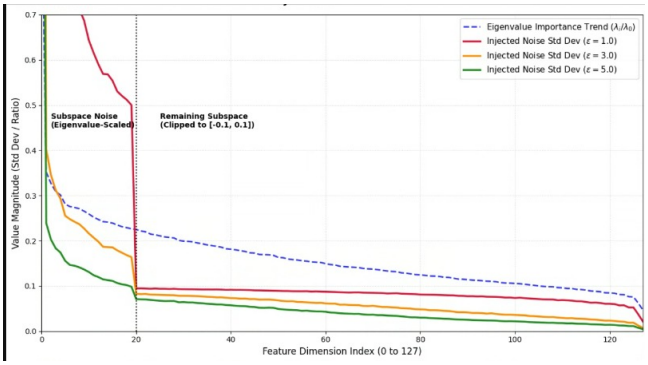


Figure 3: Empirical distribution of injected noise values across the Fisher directions on FEI dataset.

3.2.4 Template Normalization. The protected embeddings generated from the hybrid perturbation deiate form the unit hypersphere on which ArcFace embeddings are originally normalized. To preserve compatibility with traditional cosine-similarity comparisons, the perturbed embedding is projected back onto the unit hypersphere through L_2 normalization,

$$\hat{\mathbf{z}} = \frac{\tilde{\mathbf{z}}}{\|\tilde{\mathbf{z}}\|_2}, \tag{13}$$

where $\tilde{\mathbf{z}}$ denotes the perturbed embedding obtained from the proposed hybrid perturbation approach. The normalized embedding $\hat{\mathbf{z}}$ constitutes the final protected biometric template stored in the database and used for enrollment and verification.

4 Experiments and Results

This section evaluates the proposed GeoDP framework from three complementary perspectives: (i) biometric recognition performance, (ii) privacy preservation against representative attack models, and (iii) the resulting privacy-utility trade-off.

4.1 Experimental Setup

Here, we discuss the datasets, implementation details, baseline methods, and evaluation protocols used to evaluate the proposed GeoDP framework.

4.1.1 Datasets. The offline geometry learning phase requires a large and diverse collection of face datasets to estimate the discriminative structure of the ArcFace embedding space. The datasets used for offline geometry learning and online evaluation are summarized in Table 1. The offline phase uses approximately 452,000 face images from 8,100 identities from CASIA-WebFace [49] and VGGFace2 [4]. The online evaluation is conducted on four independent benchmarks. We employ the disjoint test partitions of CASIA-WebFace and VGGFace2 to evaluate in-distribution generalization, while LFW [24] and the FEI Face Database [43] are used to assess cross-dataset generalization under unconstrained and controlled imaging conditions, respectively. The identities used during offline geometry learning are disjoint from those used for evaluation.

4.1.2 Implementation Details. All experiments are conducted on a single NVIDIA Tesla T4 GPU. During the training phase, the network parameters are optimized utilizing the AdamW optimizer with an initial learning rate of 1×10^{-3} and a weight decay of 1×10^{-4}. The learning rate is modulated using a Cosine Annealing schedule over a total of 300 epochs. Input images are strictly center-cropped and resized to 112×112 pixels to process them using the pre-trained ArcFace network, extracting 512-dimensional face embeddings.

We fix δ to 10^{-5} and $\Delta_2 = 2$ following common practice for approximate differential privacy [15]. The remaining parameters (,,) are selected using the proposed Pareto optimization framework.

Table 1: Datasets used for offline geometry learning and online evaluation.

Phase	Dataset	Identities	Images
Offline	CASIA-WebFace [49]	8,000	~376,000
	VGGFace2 [4]	100	76,000
Online	CASIA-WebFace [49]	2,571	~114,000
	VGGFace2 [4]	380	~100,000
	LFW [24]	1,680	9,164
	FEI [43]	199	1,791

4.1.3 Evaluation Metrics. The proposed framework is evaluated from three perspectives.

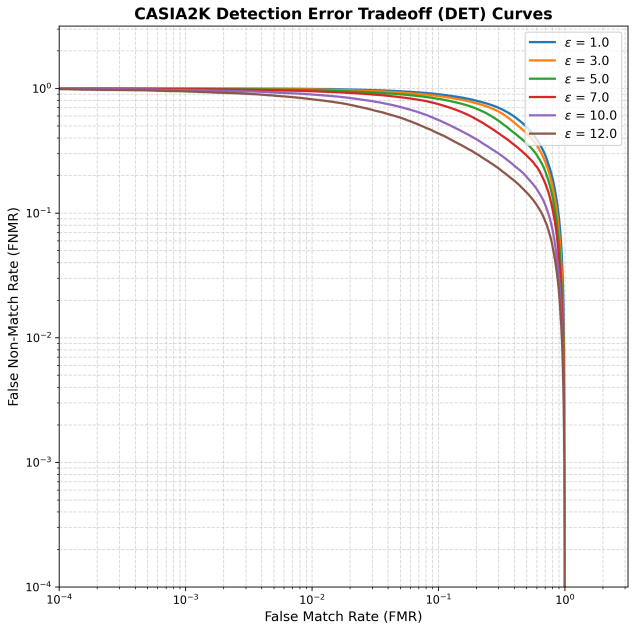
- **Recognition Performance:** Recognition utility is evaluated using verification and Rank$-k$ identification accuracy.
- **Privacy-Utility Trade-off:** The unlinkability analysis metrics (D_{sys}) and the Rank-1 accuracy threshold τ are jointly varied to obtain the empirical Pareto frontier between system privacy and utility. The selected operating points denoted as $\theta = (\epsilon, \gamma, \kappa, \alpha)$ are used throughout the experiments.
- **Security Evaluation:** The robustness of the protected templates is evaluated against template reconstruction, identity linkage, and membership inference attacks under a white-box threat model.

4.2 Recognition Performance of the Proposed GeoDP Framework

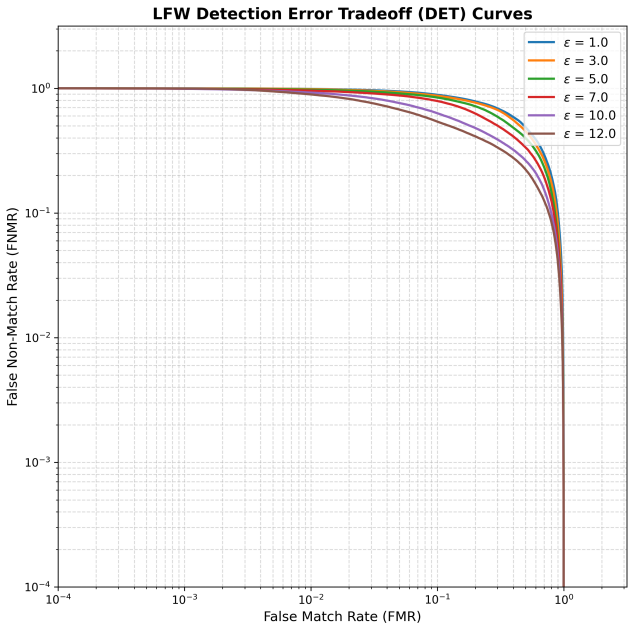
We evaluate the recognition performance of GeoDP for several values of privacy budgets (ϵ). Experiments are conducted on the datasets mentioned in Table 1 corresponding to online phase.

Table 2 summarizes the recognition performance obtained from our proposed approach. Overall, the results demonstrate the expected privacy-utility trade-off inherent to differential privacy. As the privacy budget increases from $\epsilon = 1$ to $\epsilon = 12$, the magnitude of the injected noise decreases, resulting in progressively improved recognition performance. The following are the observations drawn.

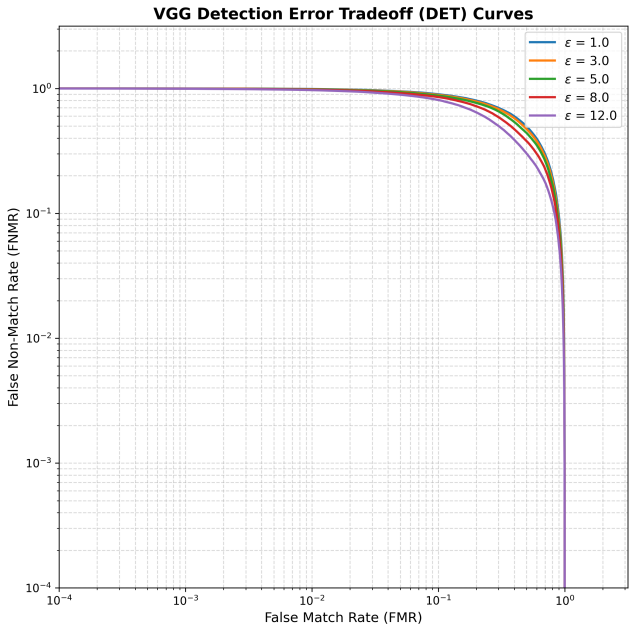
- Recognition performance under strong privacy constraints: Smaller values of ($\epsilon \leq 5$) provide stronger privacy but may substantially reduce recognition performance. However, this trade-off can be practically calibrated based on application requirements. It allows deployments to select a desired ϵ according to the target balance between privacy and recognition utility.



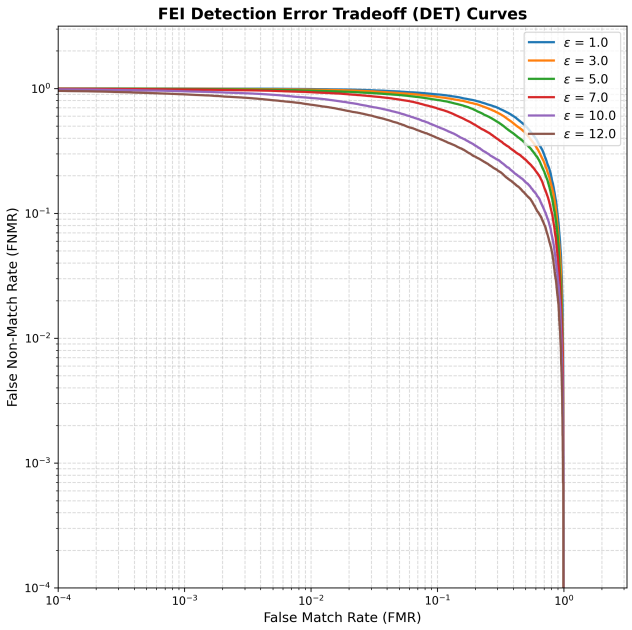
(a) CASIA-WebFace



(b) Labeled Faces in the Wild (LFW)



(c) VGGFace2



(d) FEI Face Database

Figure 4: DET curves evaluated for four datasets.

- Transition region: In general, recognition performance improves rapidly for ϵ values between 5 and 7, indicating an effective operating region where GeoDP achieves a favorable balance between privacy preservation and biometric utility. For example, the Rank-1 identification accuracy on CASIA-WebFace increases from 32.26% at $\epsilon = 3$ to 87.30% at $\epsilon = 7$; at the same time it achieves $D_{sys} = 0.3352$ demonstrating high unlinkability.

Table 2: Recognition Performance of our proposed approach on various datasets for different values of ϵ. Here, we consider $\gamma = 2, k = 20, \alpha = 0.1$.

Dataset	Privacy Budget ϵ	Verification 1:1	Identification 1:N		
		TPR @ 0.1%	Rank-1 (%)	Rank-5 (%)	Rank-10 (%)
CASIA	1	-	0.43	1.72	3.10
	3	-	32.26	48.17	54.66
	5	-	76.51	83.41	85.43
	7	-	87.30	90.27	91.19
	9	-	90.67	93.14	93.67
	12	-	92.65	94.02	94.42
LFW	1	-		0.49	1.03
	3	-		0.46	0.98
	5	-	0.08	89.72	90.60
	7	-	0.10	71.49	77.64
	9	-	0.13	76.62	81.89
	12	-	66.83	80.19	84.79
VGG	1	-	0.60	2.65	4.96
	3	-	10.30	25.21	34.99
	5	-	37.54	58.31	66.72
	7	-	53.73	71.49	77.64
	9	-	61.42	76.62	81.89
	12	-	66.83	80.19	84.79
FEI	1	-	1.05	3.98	7.13
	3	-	28.26	44.96	52.71
	5	-	67.53	74.66	78.15
	7	-	79.03	84.29	86.00
	9	-	85.34	89.82	91.70
	12	-	90.15	93.69	94.80

- Performance under relaxed privacy budgets: For $\epsilon \geq 10$, the proposed geometry-aware perturbation introduces relatively small noise while maintaining template protection. Under these settings, CASIA-WebFace and FEI achieve Rank-1 identification accuracies of 92.65% and 90.15%, respectively, demonstrating that GeoDP preserves high recognition performance at moderate privacy levels.
- Cross-dataset robustness: Recognition performance on the more challenging unconstrained datasets (VGGFace2 and LFW) remains lower than that observed on CASIA-WebFace and FEI, though Rank-10 accuracy is 84.79% for VGGFace2 and 67.15% for LFW dataset. It is because of larger intra-class variability, unconstrained imaging conditions, and significant pose and illumination changes. Nevertheless, GeoDP consistently maintains competitive recognition performance across all benchmarks, indicating that the learned geometry generalizes effectively to previously unseen identities and adverse conditions.
- We further evaluate verification results where $1:1$ comparisons are performed between the embeddings. Figure 4 shows the DET curves plotted for all 4 datasets. As observed, verification accuracy remains high for CASIA-WebFace and FEI datasets.

4.3 Comparison with Existing Approaches

We compare the proposed GeoDP framework with state-of-the-art differential privacy-based face template protection approaches under identical privacy budgets (ϵ). The evaluated baselines include:

- **PEEP: Isotropic, Gaussian noise on Face embedding [6, 7]:** Projects face images onto a PCA (Eigenface) subspace and perturbs the resulting embeddings using uniform, isotropic noise without considering the discriminative importance of individual feature directions.
- **Adaptive PCA Image Perturbation (APIP):** We construct an image-space baseline inspired by Ranked Differential Privacy (RDP) [35]. The input image is first projected onto a PCA subspace, where the principal components are ranked according to their corresponding eigenvalues. We consider the top 20 dimensions for perturbation, and thus generate 20 Gaussian noise values which are added according to the relative importance of each principal component.

Table 3 summarizes the findings given below.

- Both PCA-based perturbation and isotropic embedding perturbation result in near-random recognition performance for the given privacy budgets. The PCA-based approach perturbs the top-ranked components with Gaussian noise of decreasing magnitude, whereas isotropic perturbation adds equal noise across all embedding dimensions without considering their relative importance. These results show that neither approach alone provides a good privacy-utility trade-off.
- GeoDP consistently outperforms embedding-space isotropic Gaussian DP across all datasets. It shows that allocating noise according to the discriminative importance of the Fisher directions preserves substantially more identity information than uniformly perturbing every embedding dimension.
- Even under stringent privacy constraints ($\epsilon = 3$), GeoDP substantially outperforms both image-domain and isotropic embedding-domain baselines. As the privacy budget increases, the recognition performance progressively approaches that of the original unprotected embeddings, demonstrating that the proposed geometry-aware perturbation scales effectively across different operating conditions.
- The PCA-based baselines show high performance on LFW than on the other evaluation datasets. One possible reason may be that PCA dimensionality reduction may suppress low-variance directions that are more prominent in LFW, thus achieving a more stable representation before perturbation; though LFW remains the most challenging dataset.

4.4 Privacy Analysis

We evaluate the practical robustness of GeoDP against representative attacks that target protected biometric templates, including membership inference, template reconstruction, and cross-database linkage.

(1) **Membership Inference Attacks (MIA):** In this attack scenario, an adversary attempts to determine whether an individual's biometric data was included in the model's training set (CASIA Members) or if it is entirely unseen (Non-Members). The leakage of such information is quantified through similarity score distributions separating training set members from unseen non-members [42, 48]. We plot ROC curve that determines if an area under curve value approaches to 0.5, it states that the member and non-member template distribution are statistically indistinguishable to an

Table 3: Rank-1 Identification Accuracy (%) Comparison with Baseline and Existing Protection Approaches

Method Paradigm	Rank-1 Accuracy (%)			
	CASIA	LFW	VGGFace2	FEI
Unprotected Raw Embeddings (ArcFace [13])	96.20	82.67	84.60	99.60
Unprotected Fisherspace Projection (PCA + LDA [43])	95.87	78.47	80.30	99.33
PEEP: Isotropic noise, embedding domain (ϵ_3) [7]	0.10	0.77	0.29	0.27
PEEP: Isotropic noise, embedding domain (ϵ_{12}) [7]	0.14	0.84	0.32	0.44
APIP: Adaptive noise, image domain (ϵ_3) [35]	0.12	0.42	0.35	0.44
APIP: Adaptive noise, image domain (ϵ_{12}) [35]	1.90	2.29	1.23	1.95
Proposed Framework (ϵ_3)	32.26		10.30	28.26
Proposed Framework (ϵ_{12})	**92.65**		**66.83**	**90.15**

Table 4: Pareto-Optimal Operating Points Obtained from Multi-Objective Privacy–Utility Optimization for our proposed approach

Dataset	Pareto Parameter $\theta = (\epsilon, \gamma, \kappa, \alpha)$	Verification 1:1	Identification 1:N		
		TPR @ 0.1%	Rank-1 (%)	Rank-5 (%)	Rank-10 (%)
CASIA	(1.4, 3, 10, 0.30)	-	75.16	82.11	84.23
	(1.6, 3, 20, 0.20)	-	81.14	86.21	87.72
	(2, 3, 10, 0.10)	-	86.71	89.72	90.60
	(3, 3, 10, 0.10)	-	90.75	92.40	92.88
	(12, 3, 10, 0.10)	-	93.67	94.62	94.87
LFW	(12.0, 3, 10, 0.10)	-	61.19	67.16	69.05
VGG	(10.0, 3, 10, 0.10)	-	70.43	82.39	86.28
	(12.0, 3.0, 10, 0.10)	-	71.09	82.89	86.88
FEI	(2, 3, 10, 0.20)	-	75.05	79.81	81.52
	(3, 3, 10, 0.10)	-	80.53	84.84	86.39
	(6, 3, 10, 0.10)	-	85.40	89.54	90.59
	(11, 3, 10, 0.10)	-	90.32	93.30	94.74
	(12, 3, 10, 0.10)	-	91.75	94.46	95.63

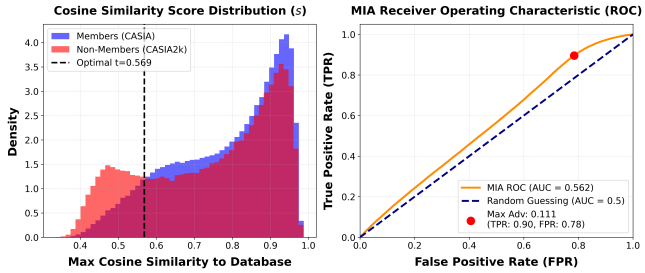
attacker. The empirical outcomes of this adversarial evaluation are shown in Figure 5.

The adversary's AUC remains marginal, peaking at only 0.567 on LFW database and 0.547 on VGGFace2. Furthermore, the Maximum Privacy Advantage achieved by the attacker across all databases is evaluated as ($Adv \implies TPR - FPR$) and is negligible (~ 0.11), heavily penalized by a False Positive Rate (FPR) exceeding 76%. This indicates that the proposed transformation effectively masks the training data.

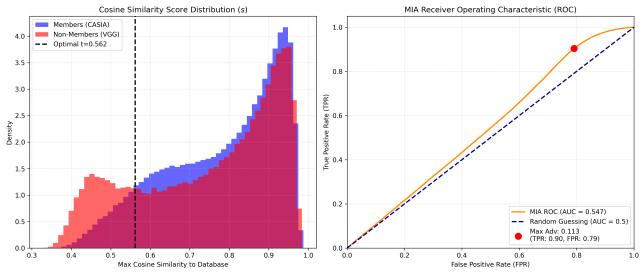
(2) **Reconstruction Attack:** In a white-box reconstruction attack, an adversary uses known original, unprotected images and corresponding protected embeddings with an inversion network (guided by L_1 and perceptual losses) to map the protected embeddings back into the visual domain, attempting to recover the user's original face structure[21, 31]. For the CASIA, LFW, and VGGFace2 datasets, the inversion attacks are executed using the NBNet architecture [10] at a privacy budget of $\epsilon = 12$. For the FEI data set, we utilize a simplified NBNet architecture customized to fit the smaller data set size and prevent generative overfitting. The structural and pixel-level distortion of the reconstructed faces relative to the ground truth images are quantified using Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index Measure (SSIM) [35, 44]. Furthermore, the residual deep identity leakage is verified by calculating the Cosine Similarity between the ArcFace embeddings of the reconstructed and original target images.

Following are our observations from Figure 6 and 7.

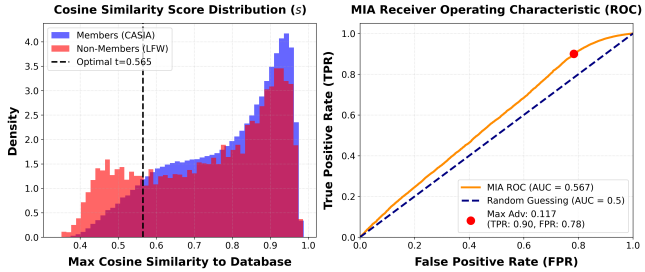
- Our framework severely limits the adversary's capacity to regenerate identifiable features even at the upper privacy budget bound ($\epsilon = 12$). For the LFW and CASIA datasets, the average SSIM score is poor ≈ 0.35 and the peak Signal-to-Noise Ratio (PSNR) does not exceed 18.0, indicating a greater distortion of the reconstruction.
- A multi-budget ablation on the FEI database at $\epsilon = 3$, $\epsilon = 5$, and $\epsilon = 12$ is shown in Figure 7. Predictably, under the strict theoretical bound of $\epsilon = 1$, the inversion network fails to recover basic facial topology. However, even as the privacy budget is relaxed to $\epsilon = 12$, the generated faces remain largely unidentifiable.



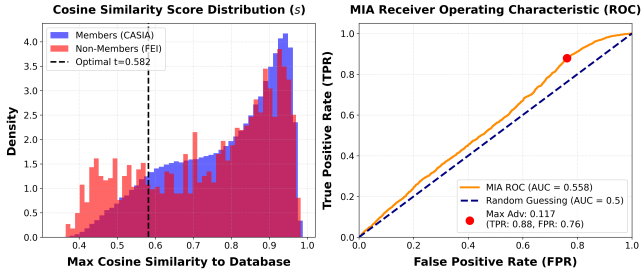
(a) Unseen CASIA (Intra-domain)



(b) VGGFace2 (Cross-domain)



(c) Labeled Faces in the Wild (LFW)



(d) FEI Face Database

Figure 5: Membership Inference Attack (MIA) evaluation. The left side displays the overlapping cosine similarity distributions between training members and unseen non-members. The right side displays the MIA ROC curves, demonstrating that the attack performance (AUC ≈ 0.55) is nearly identical to random guessing across all tested datasets.



(a) Reconstruction on LFW ($\epsilon = 12$)



(b) Reconstruction on CASIA ($\epsilon = 12$)

(c) Reconstruction on VGGFace2 ($\epsilon = 12$)

Figure 6: White-box reconstruction attack using NBNet [10]. The ground truth original faces (top rows) are compared against the adversary's reconstructions (bottom rows). The extremely low ArcFace similarity scores and SSIM degradation confirm that identity-sensitive geometry substantially degrades identity reconstruction.

- Across all ϵ configurations, the ArcFace similarity scores between the ground truth and the reconstructed variants remain well below standard matching thresholds (e.g., heavily clustered between 0.06 and 0.30). Because the high-variance principal components are subjected to concentrated Gaussian perturbation, the generative architecture remains protected.

(3) **Linkage Attacks:** To empirically validate the resistance of our proposed framework against cross-database linkage attacks, we analyze the global System Linkability metric (D_{sys}) [23, 25, 34]. It evaluates the overlap between mated (templates belonging to the same identity) and non-mated (templates from different identities) distributions within the protected embedding space stored across different systems or applications.

Figure 8 illustrates the cosine similarity score distributions across all four datasets evaluated at three distinct privacy budgets ($\epsilon \in \{3.0, 5.0, 12.0\}$). The empirical results demonstrate a clear privacy-utility-unlinkability trade-off:

- Strict Privacy ($\epsilon = 3.0$): Under stringent privacy constraints, the injected noise strictly dominates the feature space, resulting in almost near to complete overlap between mated and non-mated distributions. Consequently, D_{sys} is minimal across all datasets (e.g., 0.06 on LFW and 0.05 on VGGFace2).



Figure 7: Multi-budget reconstruction attack study on the FEI dataset using a simplified NBNet. Even as the privacy budget is relaxed from $\epsilon = 3$ to $\epsilon = 12$ to optimize utility, the identity-specific geometry remains remains difficult to recover under the evaluated attack.

- Moderate Privacy ($\epsilon = 5.0$): As the privacy budget relaxes, a controlled separation between the distributions emerges. However,linkability remains low ($D_{sys} = 0.1364$ on FEI and 0.1743 on CASIA).
- Relaxed Privacy ($\epsilon = 12.0$): At the upper boundary, the framework prioritizes utility, naturally increasing global linkability (peaking at $D_{sys} = 0.6638$ on CASIA). Though the distributions still exhibit partial overlap, preventing deterministic, zero-error template linkage even under relaxed boundary conditions.

4.5 Pareto Frontier and Privacy-Utility Optimization

To systematically map the multi-objective optimization landscape of our framework, we analyze the empirical Pareto frontiers illustrating the direct trade-off between biometric utility (Rank-1 Identification Accuracy) and privacy risk (System Linkability, denoted as D_{sys}). The hyperparameter configuration space $\theta = (\epsilon, \gamma, \kappa, \alpha)$ is densely sampled via grid search across all four benchmark databases. In Table 4, we show the optimal recognition performance obtained using the Pareto points. The target Rank-1 accuracy configurations are selected as (70%, 75%, 80%, 85%, 90%, 95%). We report only the feasible Pareto-optimal parameter configurations corresponding to the prescribed target recognition thresholds. Each configuration minimizes system linkability while satisfying the desired recognition performance, and therefore represents a privacy-utility operating point.

Figure 9 visualizes the distribution of all sampled configurations alongside the computed non-dominated Pareto optimal frontiers. Across all datasets, the framework exhibits a highly desirable trajectory, characterized by a prolonged horizontal region where biometric utility increases significantly while the system linkability risk remains tightly bounded near zero ($D_{\text{sys}} \approx 0$).

A fine-grained analysis of individual datasets highlights patterns of GeoDP.

- Intra-Domain Benchmarks (CASIA & VGGFace2): As observed in Figure 9a and Fig. 9b, the framework achieves exceptional robustness on these high-capacity distributions. For CASIA, the privacy risk remains negligible ($D_{\text{sys}} \leq 0.06$) up to an accuracy of $\tau = 90\%$, with a privacy budget of $\epsilon = 3.0$. Similar pattern is observed in VGGFace2 (Figure 9b), where the optimal configuration for $\tau = 80\%$ maintains privacy ($D_{\text{sys}} \approx 0.02$ at $\epsilon = 3.0$).
- Cross-Domain Benchmarks (LFW & FEI): Under zero-shot testing conditions on LFW (Fig. 9c), the model exhibits a compact frontier profile where to achieve maximize recognition accuracy up to $\tau = 95\%$, $\epsilon = 12.0$ is required, limiting linkability to 0.31. Conversely, the FEI database (Figure 9d) demonstrates a multi-stage trade-off. The target threshold steps through $\tau = 70\%$ ($\epsilon = 2$), $\tau = 80\%$ ($\epsilon = 3$), and ultimately $\tau = 95\%$ ($\epsilon = 12$, with $D_{\text{sys}} \approx 0.53$).

5 Security Analysis

Theorem 5.1 (Privacy Preservation and Template Indistinguishability of GeoDP). *Let the Fisher projection matrix and the geometry-aware covariance matrix be learned offline from a representative population and remain fixed during enrollment and authentication. Then, the proposed GeoDP framework satisfies (ϵ, δ)-differential privacy.*

Proof. Let $\mathbf{x} \in \mathbb{R}^d$ denote the original face embedding and $\mathbf{z} = V^T\mathbf{x}$, where V is the Fisher projection matrix learned during the offline training stage. GeoDP first generates a perturbed embedding

$$\mathbf{z}' = \mathbf{z} + \mathbf{n},$$

where $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \Sigma)$, and the geometry-aware covariance matrix is $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_d^2)$, with

$$\sigma_i = \sigma_0 \left(\frac{\lambda_i}{\lambda_1}\right)^{\gamma}, \qquad \sigma_0 = \frac{\Delta_2\sqrt{2\ln(1.25/\delta)}}{\epsilon}.$$

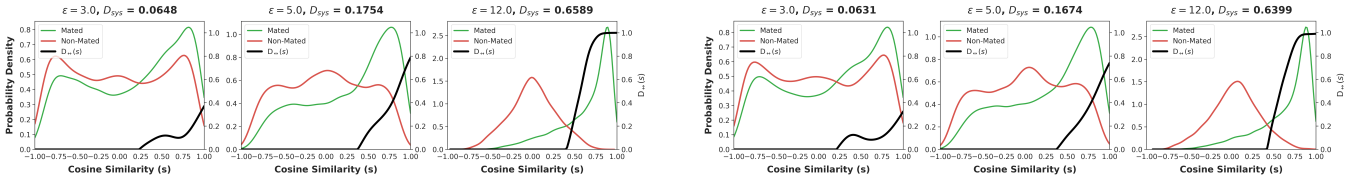
Since V, $\{\lambda_i\}$, and Σ are learned offline and remain fixed for all users, the perturbation is a multivariate Gaussian mechanism which is independent of the input biometric template used in the online phase. Therefore, for any pair of neighboring embeddings $\mathbf{x}$ and $\mathbf{x}'$, and for any measurable output set $\mathcal{S}$,

$$\Pr[\mathcal{M}_G(\mathbf{x}) \in \mathcal{S}] \leq e^{\epsilon} \Pr[\mathcal{M}_G(\mathbf{x}') \in \mathcal{S}] + \delta,$$

where $\mathcal{M}_G(\cdot)$ denotes the geometry-aware Gaussian perturbation mechanism.

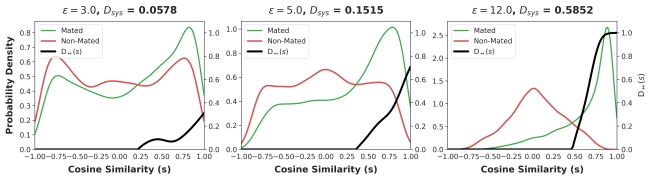
Next, GeoDP applies two deterministic operations- clipping only in weakly discriminative directions and ℓ_2 normalization, together denoted as

$$T(\mathbf{z}') = \frac{\text{clip}(\mathbf{z}', -\alpha, \alpha)}{\|\text{clip}(\mathbf{z}', -\alpha, \alpha)\|_2}.$$

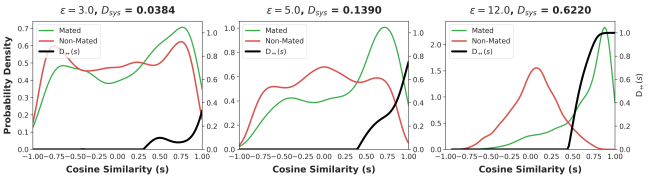


(a) CASIA-WebFace

(b) Labeled Faces in the Wild (LFW)

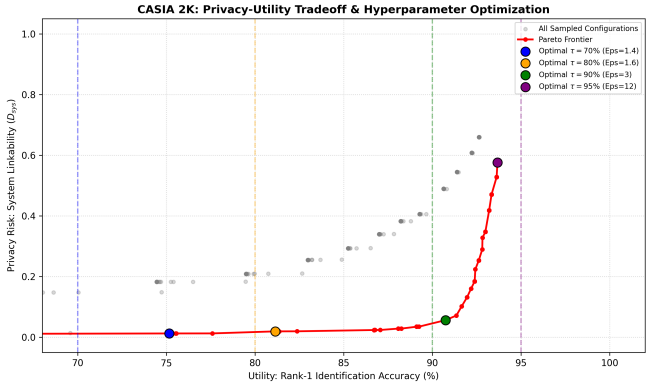


(c) VGGFace2

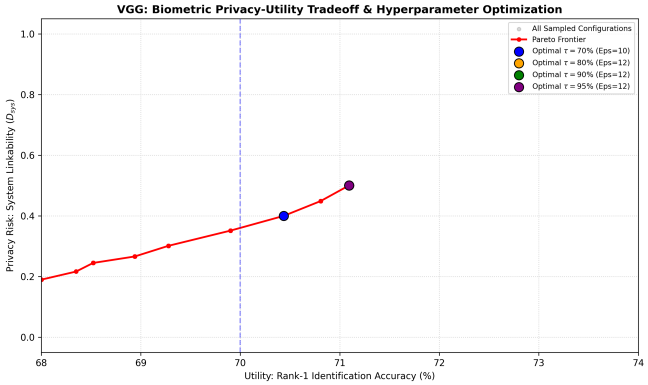


(d) FEI Face Database

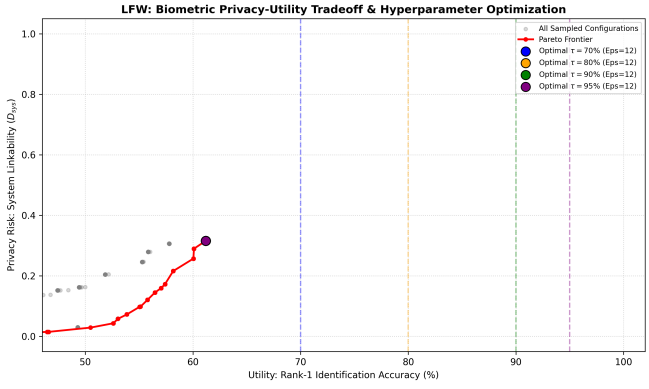
Figure 8: System Linkability (D_{sys}) evaluation across four datasets.



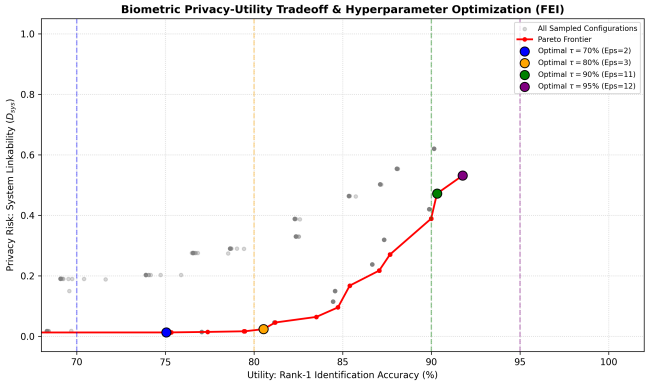
(a) CASIA-WebFace Test Split



(b) VGGFace2 Test Split



(c) Labeled Faces in the Wild (LFW)



(d) FEI Face Database

Figure 9: Biometric privacy-utility trade-off curves and multi-variable hyperparameter optimization across four distinct databases. The red line traces the non-dominated Pareto frontier. Optimal operational points are highlighted for targeted recognition thresholds ($\tau \in \{70\%, 80\%, 90\%, 95\%\}$) alongside their required minimum privacy budgets (ϵ).

The complete GeoDP mechanism can be denoted $\mathcal{M}_{GeoDP}(\mathbf{x}) = T(\mathcal{M}_G(\mathbf{x}))$. These deterministic functions are applied on the already perturbed embedding and they do not depend on the original biometric template, thus cannot increase information leakage. Hence,

$$\Pr[\mathcal{M}_{GeoDP}(\mathbf{x}) \in \mathcal{S}] \leq e^{\epsilon} \Pr[\mathcal{M}_{GeoDP}(\mathbf{x}') \in \mathcal{S}] + \delta. \tag{14}$$

We consider an adversary that receives a protected template generated from either $\mathbf{x}_0$ or $\mathbf{x}_1$, where the choice is represented by a random bit $b \in \{0, 1\}$. Using Eq. 14, even when the adversary knows the complete GeoDP algorithm, and all public parameters, it cannot determine correctly whether the protected template is generated from $\mathbf{x}_0$ or $\mathbf{x}_1$ beyond the distinguishability allowed by (ϵ, δ).

Hence, GeoDP provides both (ϵ, δ)-differential privacy and template indistinguishability under the white-box threat model. □

5.1 Known Auxiliary-Template Attack

The above theorem considers the Protected-Template-Only Attack. We consider a stronger adversary that possesses auxiliary pairs of original and protected embeddings,

$$\mathcal{D} = \{(\mathbf{x}_i, \tilde{\mathbf{x}}_i)\}_{i=1}^{q}, \qquad \tilde{\mathbf{x}}_i = \mathcal{M}_{GeoDP}(\mathbf{x}_i).$$

Such auxiliary observations can be used to learn the statistical behavior of the protection algorithm or to train an inversion model (as discussed in Section 4). However, for each execution of the GeoDP algorithm, we use the independent perturbation randomness. Therefore, the auxiliary input-output pairs do not reveal the specific perturbation/noise value used for an independently protected template. Similar to the above theorem, in the presence of known auxiliary pairs, the distinguishing guarantee remains bounded by the privacy parameters.

6 Security and Privacy Analysis

We analyze the privacy and security properties of GeoDP under a white-box threat model, where the adversary is assumed to know the complete protection mechanism and all its parameters, but not the random perturbation sampled for a particular template instance.

6.1 Privacy Properties of GeoDP

Let $\mathbf{x} \in \mathbb{R}^d$ denote an input face embedding. GeoDP first projects the embedding into the learned Fisher subspace as

$$\mathbf{z} = V^T \mathbf{x}, \tag{15}$$

where V is learned offline from a representative population and remains fixed during deployment. Geometry-aware perturbation is then generated independently for each Fisher direction as

$$n_i \sim \mathcal{N}(0, \sigma_i^2), \qquad \sigma_i = \sigma_0 \left(\frac{\lambda_i}{\lambda_1}\right)^{\gamma}, \tag{16}$$

where $\{\lambda_i\}$ denotes the Fisher eigenspectrum and γ controls the allocation of perturbation across the learned directions. The hybrid perturbation mechanism is defined as

$$\tilde{z}_i = \begin{cases} z_i + n_i, & i \leq k, \\ z_i + \text{clip}(n_i, -\alpha, \alpha), & i > k, \end{cases} \tag{17}$$

followed by ℓ_2 normalization,

$$\hat{\mathbf{z}} = \frac{\tilde{\mathbf{z}}}{\|\tilde{\mathbf{z}}\|_2}. \tag{18}$$

The parameters V, $\{\lambda_i\}$, γ, k, and α are determined during the offline design stage and remain fixed during enrollment and authentication. Thus, the protection rule itself does not depend on the identity of the input user. However, the perturbation $\mathbf{n}$ is independently sampled for every execution of GeoDP.

The underlying perturbation is based on a geometry-aware Gaussian mechanism, with the noise scale controlled by the selected privacy budget. In contrast to isotropic mechanisms, the covariance is adapted according to the discriminative structure of the Fisher subspace. The hybrid component further modifies the perturbation in the weakly discriminative directions by limiting large noise realizations through the threshold α. Consequently, the hybrid mechanism should be viewed as a controlled modification of the underlying geometry-aware Gaussian perturbation rather than an identical standard Gaussian mechanism.

For two candidate embeddings $\mathbf{x}_0$ and $\mathbf{x}_1$, the corresponding protected templates are random variables,

$$\hat{\mathbf{z}}_b = \mathcal{M}_{\text{GeoDP}}(\mathbf{x}_b), \qquad b \in \{0, 1\}. \tag{19}$$

Since fresh perturbation is sampled independently for each execution,

$$\mathcal{M}_{\text{GeoDP}}(\mathbf{x}; \mathbf{n}_1) \neq \mathcal{M}_{\text{GeoDP}}(\mathbf{x}; \mathbf{n}_2), \qquad \mathbf{n}_1 \neq \mathbf{n}_2, \tag{20}$$

in general. Therefore, GeoDP does not define a deterministic mapping between an original embedding and its protected representation. Even when the adversary knows V, $\{\lambda_i\}$, γ, k, and α, the particular perturbation realization used for a target template remains unknown.

The final ℓ_2 normalization is a deterministic transformation applied after the randomized perturbation stage. Hence, it does not introduce additional information about the original embedding beyond that contained in the perturbed representation.

6.2 Protected-Template and Auxiliary-Template Attacks

We first consider a protected-template-only adversary that observes a protected template

$$\hat{\mathbf{z}} = \mathcal{M}_{\text{GeoDP}}(\mathbf{x}), \tag{21}$$

and has complete knowledge of the GeoDP mechanism. Given two candidate embeddings $\mathbf{x}_0$ and $\mathbf{x}_1$, the adversary attempts to determine the source embedding corresponding to the observed protected template. Since the mechanism incorporates independently sampled randomness, the conditional output distribution

$$p(\hat{\mathbf{z}} \mid \mathbf{x}) \tag{22}$$

is non-degenerate and varies according to the geometry-aware and hybrid perturbation mechanism. Consequently, knowledge of

the protection algorithm does not provide a deterministic inverse mapping from $\hat{\mathbf{z}}$ to $\mathbf{x}$.

We next consider a stronger auxiliary-template adversary with access to a set of original–protected template pairs,

$$\mathcal{D} = \{(\mathbf{x}_i, \hat{\mathbf{z}}_i)\}_{i=1}^{q}, \qquad \hat{\mathbf{z}}_i = \mathcal{M}_{\text{GeoDP}}(\mathbf{x}_i). \tag{23}$$

The auxiliary pairs may help characterize the statistical behavior of the protection mechanism or learn an approximate mapping; however, because each execution uses independently sampled perturbation, they do not reveal a deterministic mapping or the specific perturbation realization used for a newly protected target template."

Such observations may allow the adversary to estimate the statistical behavior of the protection mechanism or train an approximate inversion model. However, because independent perturbation is sampled for every execution, the observed pairs do not reveal the particular noise realization associated with an independently protected target template. Thus, auxiliary information may characterize the distribution of the mechanism but does not establish a deterministic mapping between an original embedding and a newly generated protected template.

The resistance of GeoDP against adversaries exploiting such auxiliary information is further evaluated through reconstruction, membership inference, and unlinkability experiments under the assumed white-box setting.

7 Related Work

Existing biometric template protection schemes are broadly classified into biometric cryptosystems, cancelable biometrics, privacy preserving approaches and deep learning based approaches. Biometric cryptosystems such as fuzzy vault [27] and fuzzy commitment [14] bind the biometric template to a cryptographic key with the help of a helper data. Though these schemes are theoretically well established, these do not handle high error rates present in the biometric templates. Cancelable biometrics [9, 22, 30, 36] perform a non-invertible transformation to the original template before storing and comparisons are performed on the transformed domain. However, they provide no formal, quantifiable privacy guarantees. Privacy preserving approaches such as those based on Homomorphic based encryption are gaining popularity. However, the above-mentioned approaches either suffer from degraded recognition performance or lack security and privacy properties. Deep learning based methods [22, 30] use a user specific key to randomize the face template before it is passed through neural network and generate the protected face template. Such learning-based approaches shows high recognition performance; however, the user derived key itself has some security concerns.

Dwork *et al.* [16] established the mathematical foundations of differential privacy (DP), defining (ε, δ)-DP. In the face biometrics domain, Mirjalili *et al.* [32] proposed PEEP (Privacy using Eigen-Face Perturbation), which applies isotropic Gaussian DP noise to PCA eigenface coefficients before storage. It adds identical noise to all PCA components regardless of their discriminative importance, wasting privacy budget on low-variance dimensions while under-protecting the most identity-sensitive ones. Ou *et al.* [35] proposed Ranked Differential Privacy (RDP), which performs multiscale wavelet decomposition of face images. It applies Laplacian noise to sparse feature coefficients weighted by their rank according to the privacy budget. The limitation is discussed in Section 4.

8 Conclusion

Differential privacy provides a principled way to protect biometric templates, but its application in face recognition without affecting performance remains challenging. Existing approaches either perturb the input image before feature extraction or add uniform noise to the embedding, without considering the statistical geometry of deep face embeddings, leading to low recognition performance. In this work, we propose *GeoDP*, a geometry-aware perturbation framework that operates directly on face embeddings. By learning a Fisher subspace and using eigenspectrum, it adapts the noise distribution to the structure of the embedding space. We further introduce a hybrid perturbation strategy that limits large noise realizations in weakly discriminative directions. Experiments on CASIA-WebFace, VGGFace2, LFW, and FEI show that the proposed approach provides a better trade-off between recognition performance and privacy than the existing approaches. At $\epsilon = 12$, GeoDP achieves 93.67% Rank-1 accuracy on CASIA-WebFace and 91.75% on FEI. The protected templates also show effective protection from reconstruction attack, membership inference attack, and provide high unlinkability under the white-box threat model. Broadly, the proposed framework highlights the idea of designing privacy mechanisms around the characteristics of learned representations. It can extend beyond face biometrics to other biometric modalities and privacy-sensitive machine learning applications, where adapting protection mechanisms to the underlying data representation may help achieve a better balance between privacy and utility.

Future work will explore the application of the proposed framework to other biometric modalities and investigate whether the geometry and perturbation mechanism can be learned jointly with the underlying representation model.

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A Research Methods

A.1 Part One

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A.2 Part Two

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B Online Resources

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